# STROKE UNIT ADMISSION CRITERIA

## Aligned with American Heart Association / American Stroke Association (AHA/ASA)

Guidelines and Stroke Systems of Care for Stroke Center Accreditation Readiness

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*Document Type: Clinical Policy / Accreditation Reference*

Applicable Programs: Acute Stroke-Ready Hospital (ASRH) · Primary Stroke Center (PSC) · Thrombectomy-Capable Stroke Center (TSC) · Comprehensive Stroke Center (CSC)

**King Fahad Central hospital — Department of Neurology / Stroke Program**
Effective Date: _____________ Review Cycle: Annual Version: 1.0
Approved by: Stroke Medical Director / Stroke Program Coordinator / Quality Committee

## 1. Purpose

This document establishes standardized admission criteria for the inpatient Stroke Unit, consistent with American Heart Association/American Stroke Association (AHA/ASA) scientific statements and guidelines on stroke systems of care, Get With The Guidelines–Stroke (GWTG-Stroke) performance measures, and the structural/process requirements referenced by national stroke center certification programs (e.g., The Joint Commission, DNV Healthcare, and state-designated stroke systems that use AHA/ASA-aligned criteria). It is intended to support survey readiness and demonstrate structured, evidence-based patient selection for stroke unit level of care.

This policy applies to all patients presenting with suspected or confirmed acute stroke (ischemic stroke, transient ischemic attack, intracerebral hemorrhage, and clinically stabilized subarachnoid hemorrhage) being considered for admission, transfer, or step-down to the designated Stroke Unit.

## 2. Regulatory and Guideline Basis

Admission criteria below are derived from the following AHA/ASA source documents and are intended to be cited in accreditation site-visit documentation:

- 2026 AHA/ASA Guideline for the Early Management of Patients With Acute Ischemic Stroke.
- Current AHA/ASA stroke systems-of-care and stroke-center development scientific statements.
- AHA/ASA Scientific Statement: Ideal Foundational Requirements for Stroke Program Development and Growth (stroke unit structural/staffing framework).
- AHA/ASA Get With The Guidelines–Stroke (GWTG-Stroke) quality measures and clinical decision pathways.
- AHA/ASA Guidelines for the Management of Spontaneous Intracerebral Hemorrhage.
- AHA/ASA Guidelines for the Management of Aneurysmal Subarachnoid Hemorrhage.
- AHA/ASA/AAN Guidelines for Prevention of Stroke in Patients With Stroke and TIA (secondary prevention pathway).

## 3. Definition of the Stroke Unit

A geographically defined inpatient nursing unit (or clearly demarcated group of beds) dedicated to the care of stroke patients, staffed by nurses and providers with documented stroke-specific competency, and equipped for continuous or intermittent neurological and cardiac monitoring, distinct from a general medical-surgical floor and from a dedicated neuro-intensive/critical care unit.

## 4. General Admission Criteria (Inclusion)

A patient is appropriate for direct admission or transfer to the Stroke Unit when ALL of the following are met:

1. Diagnosis of acute ischemic stroke, hemorrhagic stroke (post-stabilization), transient ischemic attack (TIA) at intermediate-to-high recurrence risk, or stroke mimic requiring monitored observation.
2. Hemodynamically and respiratory stable without need for vasopressors, invasive mechanical ventilation, or continuous airway support.
3. Glasgow Coma Scale (GCS) ≥ 9, or a lower GCS explained by a stable, non-progressive deficit and approved by the stroke attending.
4. Baseline NIHSS documented and reassessment schedule ordered per protocol.
5. No anticipated need for emergent neurosurgical intervention at time of admission.
6. Can be safely managed with the unit's approved nurse-to-patient ratio and monitoring capability (see Section 7).

### 4.1 Inclusion Criteria by Stroke Subtype

| Category | Admission Criteria |
|---|---|
| Acute Ischemic Stroke (AIS) | Confirmed AIS on CT/MRI or high clinical probability with negative initial imaging; post-IV thrombolysis patients after completion of the mandated post-tPA monitoring interval defined by protocol; not requiring ICU-level airway or hemodynamic support. |
| Post-Endovascular Thrombectomy (EVT) | Successful or attempted mechanical thrombectomy, hemodynamically stable, extubated (or never intubated), no evidence of procedural complication requiring ICU care; per-protocol post-EVT monitoring interval eligible for step-down to Stroke Unit. |
| Transient Ischemic Attack (TIA) | ABCD2 score ≥ 4, crescendo TIA, TIA with large-vessel/cardioembolic source suspected, or inability to complete urgent outpatient work-up within 24–48 hours. |
| Intracerebral Hemorrhage (ICH) | Small-to-moderate volume ICH, hemodynamically stable, GCS ≥ 9, controlled blood pressure per protocol, no impending herniation, and neurosurgery has confirmed no acute operative indication. |
| Subarachnoid Hemorrhage (SAH) | Only after securing of aneurysm (or determination not to intervene) and clearance by neurosurgery/neurocritical care for step-down; vasospasm monitoring capability confirmed on the unit if admitted during the risk window. |
| Stroke Mimic / Rule-out | Patients requiring monitored neurological observation and serial exams pending diagnostic clarification (e.g., complex migraine, seizure with Todd's paralysis, functional neurological disorder) at treating physician discretion. |

## 5. Exclusion Criteria (Requires ICU / Neuro-ICU Level of Care)

The following findings mandate ICU or neurocritical care admission rather than Stroke Unit admission:

- Need for invasive mechanical ventilation or inability to protect the airway.
- Hemodynamic instability requiring vasopressor or inotropic support.
- GCS < 9, or acute decline in GCS ≥ 2 points, not explained by a stable baseline deficit.
- Malignant MCA infarction or large-territory infarct at risk for cerebral edema/herniation requiring osmotic therapy titration or neurosurgical monitoring.
- Post-hemicraniectomy or other neurosurgical postoperative status requiring ICU-level monitoring.
- Uncontrolled intracranial pressure or need for an external ventricular drain / intracranial pressure monitor.
- Symptomatic hemorrhagic transformation post-thrombolysis with hemodynamic or neurological compromise.
- SAH within the active vasospasm window at institutions where the Stroke Unit is not equipped for that level of monitoring.
- Status epilepticus or refractory seizures.
- Need for continuous intra-arterial blood pressure monitoring or titrated intravenous antihypertensive/pressor infusions beyond unit-approved agents.

# 6. Structural and Staffing Requirements Supporting Admission

## 6.1 Staffing

- Dedicated stroke-competent nursing staff with documented completion of stroke-specific orientation/competency validation (e.g., NIHSS certification) within a defined interval of hire and on a recurring (typically annual) basis.
- Defined maximum nurse-to-patient ratio for the unit (institution-specific; commonly no greater than 1:3–1:4 for stroke unit acuity), distinct from general medical-surgical ratios.
- 24/7 availability of a physician or advanced practice provider with stroke expertise, either on-site or via telestroke, for emergent evaluation.
- Defined stroke team activation pathway with documented response-time targets.

## 6.2 Monitoring Capability

- Continuous cardiac (telemetry) monitoring capability for all admitted stroke patients, per AHA guidance on arrhythmia detection (e.g., atrial fibrillation) during the acute phase.
- Capability for frequent, protocol-driven neurological checks and vital sign monitoring (see Section 7).
- Pulse oximetry and, where indicated, non-invasive blood pressure monitoring at physician-ordered intervals.

## 6.3 Multidisciplinary Team

| Role | Expected Involvement |
|---|---|
| Stroke Medical Director / Neurologist | Oversight of admission criteria, protocol deviations, and quality review. |
| Stroke-Certified Registered Nurses | Primary bedside care, NIHSS reassessment, protocol-driven monitoring. |
| Rehabilitation Services (PT/OT/SLP) | Assessment within 24–48 hours of admission; dysphagia screening prior to any oral intake. |
| Pharmacy | Medication reconciliation, anticoagulation/antithrombotic management, thrombolytic protocol support. |
| Case Management / Social Work | Discharge planning, post-acute care coordination, secondary prevention follow-up scheduling. |
| Dietitian | Nutritional assessment, particularly for patients with dysphagia. |

# 7. Post-Admission Monitoring Standards

The following protocol-driven monitoring parameters should be ordered on admission and are commonly reviewed by accreditation surveyors as evidence of standardized stroke unit care:

| Parameter | Standard |
|---|---|
| Neurological checks / NIHSS | Per protocol (commonly every 1–2 hours for the first 24 hours post-thrombolysis/thrombectomy, then per physician order and unit protocol). |
| Vital signs | Per protocol intensity based on treatment received (e.g., closer intervals post-tPA/EVT, tapering per defined schedule). |
| Blood pressure parameters | Physician-ordered target range specific to treatment status (e.g., post-thrombolysis BP goals) with defined escalation triggers. |
| Glucose management | Point-of-care glucose monitoring with treatment protocol; avoidance of hyperglycemia and hypoglycemia per AHA guidance. |

| Parameter | Standard |
|---|---|
| Temperature | Monitoring and treatment of fever; source investigation if persistent. |
| Dysphagia screening | Validated bedside swallow screen prior to any oral intake, medications, or food, ideally before the patient leaves the emergency department or within the first hours of admission. |
| Cardiac monitoring | Continuous telemetry for arrhythmia detection, particularly atrial fibrillation, for a guideline-defined minimum duration. |
| VTE prophylaxis | Mechanical prophylaxis on admission; pharmacologic prophylaxis per protocol once hemorrhage risk is assessed. |
| Early mobilization | Assessment for safe, early (as tolerated) mobilization per interdisciplinary protocol. |
| Secondary prevention work-up | Initiation of etiologic work-up (echocardiogram, vascular imaging, lipid panel, HbA1c, cardiac rhythm monitoring) during admission. |

## 8. Escalation and Transfer-Out Criteria

Immediate escalation to a higher level of care (ICU, neurocritical care, or transfer to a Comprehensive Stroke Center / Thrombectomy-Capable Stroke Center) is required if any of the following occur while on the Stroke Unit:

- Decline in GCS by ≥ 2 points or new/worsening focal deficit suggestive of hemorrhagic transformation, re-occlusion, or progressive edema.
- Systolic blood pressure or other vital sign parameter exceeds physician-defined escalation threshold despite first-line intervention.
- New respiratory compromise or inability to protect the airway.
- Seizure activity.
- Identification of a large-vessel occlusion or hemorrhage requiring intervention not available at the current facility level, triggering the organization's stroke transfer protocol to a higher-level certified center.

## 9. Documentation and Quality Metrics

The following are commonly required data elements/metrics for AHA/ASA Get With The Guidelines–Stroke participation and certification survey review; the Stroke Unit admission record should support capture of each:

- Time of symptom onset / last known well.
- Door-to-CT, door-to-needle, and door-to-groin-puncture times (as applicable).
- NIHSS on arrival, post-treatment, and at discharge.
- Dysphagia screen completion prior to oral intake.
- VTE prophylaxis by end of hospital day 2.
- Antithrombotic therapy by end of hospital day 2 (when indicated).
- Statin therapy at discharge (when indicated).
- Rehabilitation assessment completion.
- Stroke education and discharge planning documentation.
- Modified Rankin Scale or equivalent functional outcome at discharge.

## 10. Policy Review and Ownership

This document is owned by the Stroke Program Coordinator and Stroke Medical Director and is reviewed at minimum annually, or upon release of updated AHA/ASA guidance or certifying-body standards, whichever occurs first. All revisions require approval by the institution's Stroke Committee / Quality Council prior to implementation.

## 11. References

American Heart Association/American Stroke Association. 2018 Guidelines for the Early Management of Patients With Acute Ischemic Stroke. Stroke. 2018;49:e46–e110 (with subsequent focused updates).

American Heart Association/American Stroke Association. Formation and Function of Acute Stroke-Ready Hospitals Within a Stroke System of Care: Scientific Statement. Stroke. 2019.

American Heart Association/American Stroke Association. Ideal Foundational Requirements for Stroke Program Development and Growth: Scientific Statement. Stroke.

American Heart Association/American Stroke Association. Get With The Guidelines–Stroke Program Manual and Measures Overview. heart.org/qualityresearch.

American Heart Association/American Stroke Association. Guidelines for the Management of Spontaneous Intracerebral Hemorrhage.

American Heart Association/American Stroke Association. Guidelines for the Management of Aneurysmal Subarachnoid Hemorrhage.

The Joint Commission. Comprehensive, Thrombectomy-Capable, Primary, and Acute Stroke-Ready certification program requirements (reference current edition used by the organization).